Antihyperlipidemic drugs

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Objectives

- บอกชื่อยา และจำแนกกลุ่มยารักษาไขมันในเลือดผิดปกติ
- อธิบาย
  - กลไกการออกฤทธิ์
  - เภสัชจลนศาสตร์
  - ผลของยาต่อระดับไขมันในเลือด
  - อาการไม่พึงประสงค์ ข้อควรระวังและข้อห้ามใช้

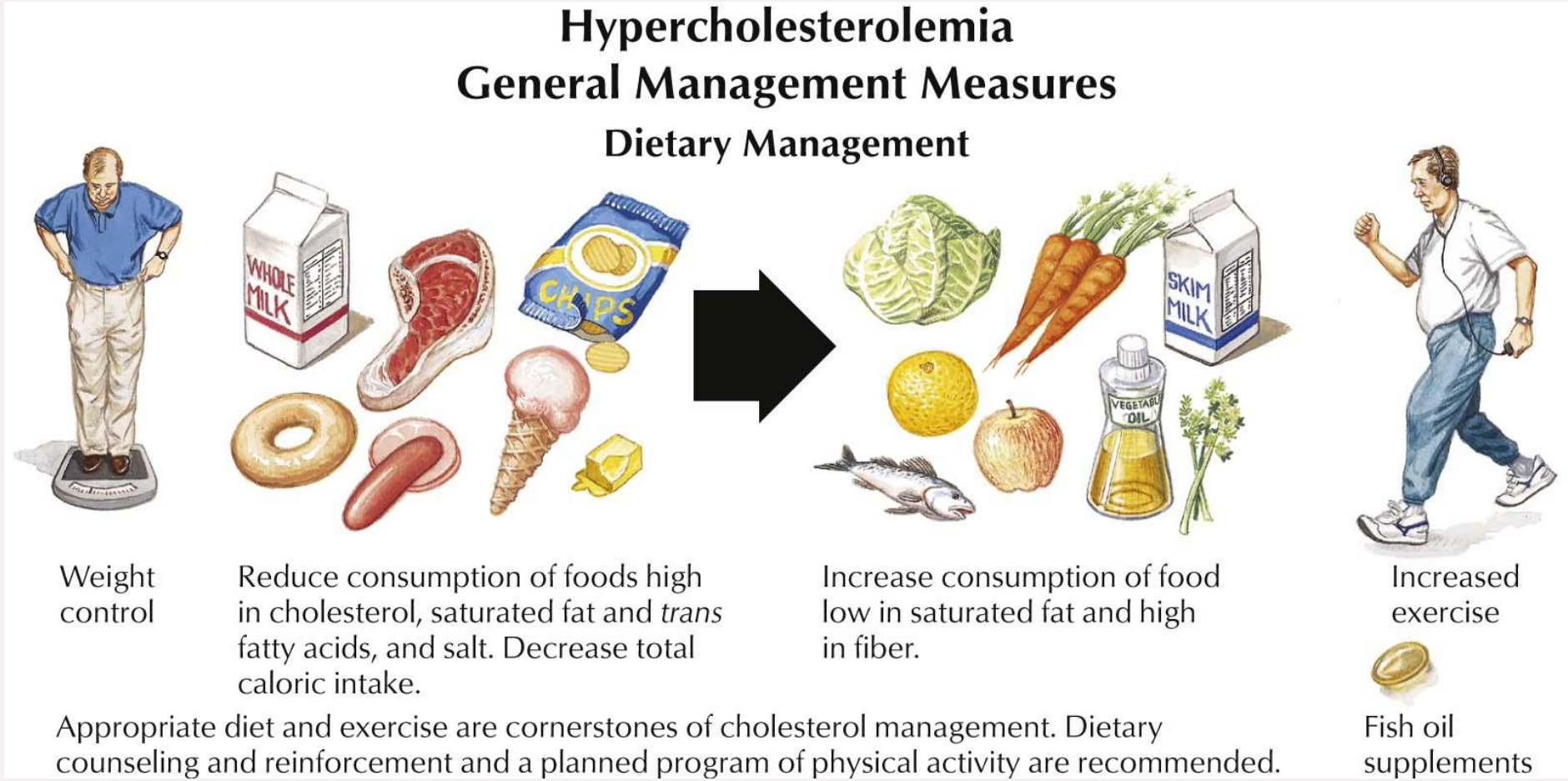
Dyslipidemia

- Disorder of lipoprotein metabolism
  - characterized by:
    - increased plasma levels of cholesterol, triglycerides, or both
    - reduced levels of HDL-C
  - increased risk for atherosclerotic cardiovascular disease (ASCVD)

Treatment of dyslipidemia

Treatment target: to reduce ASCVD risk in adults

- **Lifestyle modifications**



(Netter's Illustrated Pharmacology 2014, 2nd Edition)

Treatment of dyslipidemia

- **Drugs for treatment of dyslipidemias**

  1. **Statins** (HMG-CoA reductase inhibitors)***
  2. Cholesterol absorption inhibitors
  3. PCSK9 inhibitors
  4. Fibrates
  5. Bile acid-binding resins (bile acid sequestrants)
  6. ATP-citrate lyase inhibitor
  7. Nicotinic acid (Niacin)

- Omega-3 fatty acid
- Inhibitor of Angiopoietin-Like Protein 3: Evinacumab-dgnb (approved in 2021)

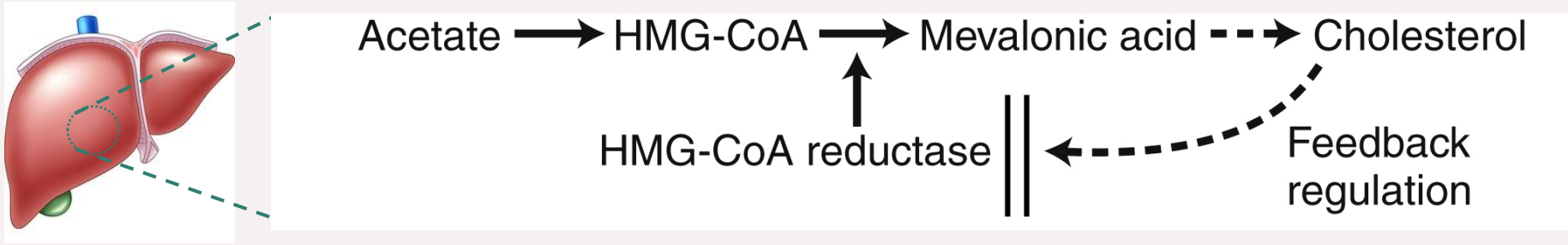
(HMG-CoA: 3-hydroxy-3-methylglutaryl coenzyme A)
(PCSK9: Proprotein convertase subtilisin/kexin type 9)

Statins

- HMG-CoA reductase inhibitors
- lovastatin (prodrug), simvastatin (prodrug), fluvastatin, pravastatin, atorvastatin, rosuvastatin, pitavastatin

HMG-CoA reductase

- rate-limiting enzyme in hepatic cholesterol biosynthesis, which convert HMG-CoA to mevalonic acid



Statins

Mechanism of action

- competitively inhibits HMG-CoA reductase

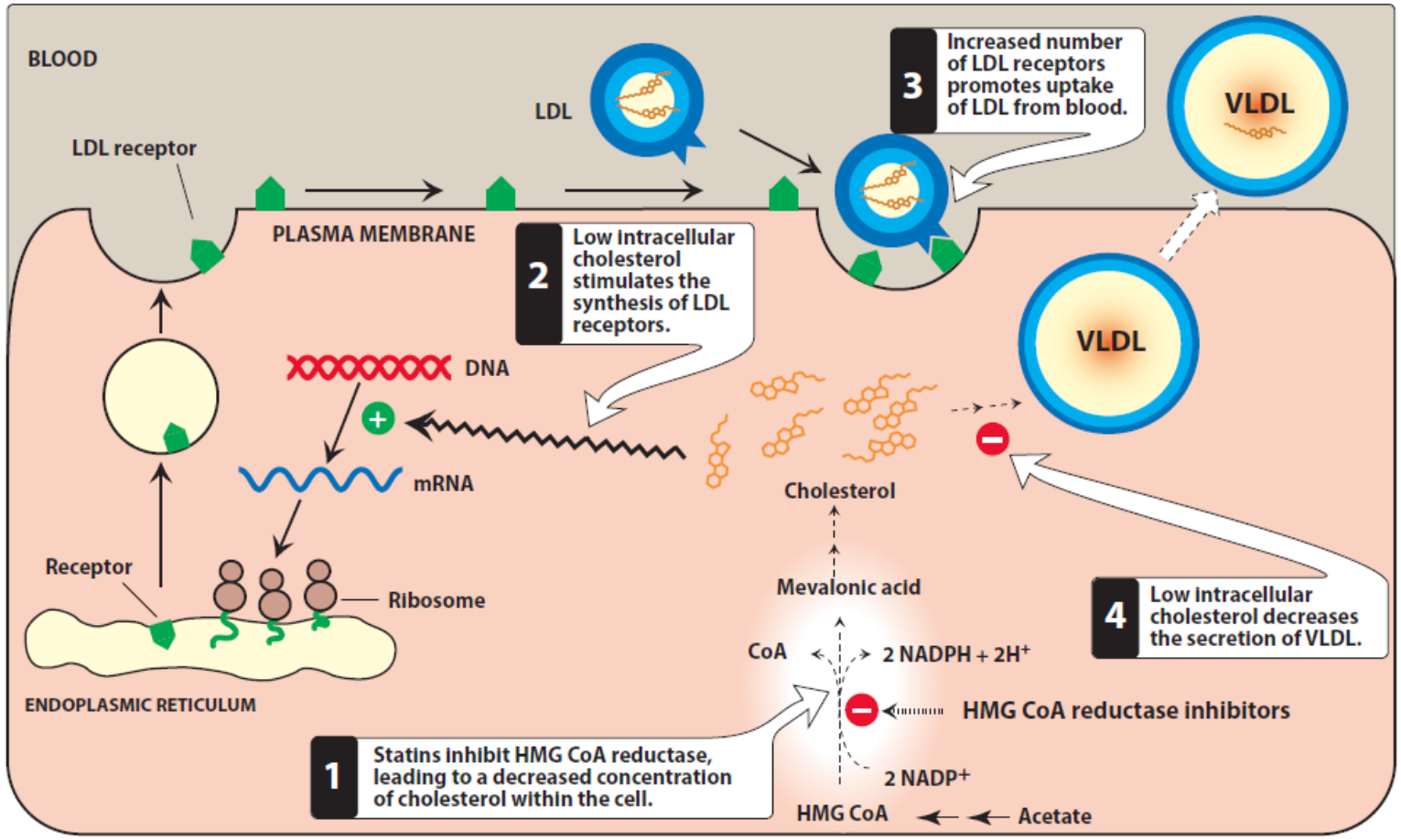


Figure 23.5
Inhibition of HMG CoA reductase by the statin drugs. HMG CoA = 3-hydroxy-3-methylglutaryl coenzyme A; LDL = low-density lipoprotein; VLDL = very–low-density lipoprotein.

- inhibit hepatic cholesterol synthesis
- increase synthesis of LDL receptor
- increase removal of LDL from blood
- decrease VLDL synthesis

Statins

Pharmacokinetics

- **Absorption**
  : occurs in the intestine (30-85%) after oral administration
- **Metabolism**
  : extensive 1st pass hepatic uptake, mediated by anion transporter **OATP1B1**
  - short plasma $t_{1/2}$ (1-3 h) -- fluvastatin, lovastatin, simvastatin, pravastatin
  - long plasma $t_{1/2}$ -- atorvastatin (14 h), pitavastatin (12 h), rosuvastatin (19 h)

  : metabolized by CYP450 enzyme system
    - CYP3A4 (lovastatin, simvastatin, atorvastatin)
    - CYP2C9 (fluvastatin)

  : sulfation-pravastatin
- **Excretion**
  : bile and feces (>70%)
  : urine (5-20%)

Statins

Effects on plasma lipoprotein levels

- ↓ **LDL** : dose dependent
- ↓ TG
- ↑ HDL

High-Intensity Statin	Moderate-Intensity Statin	Low-Intensity Statin
lowers LDL-C by ~ ≥50%	lowers LDL-C by ~ 30-<50%	lowers LDL-C by ~ <30%
-	Lovastatin 40 mg	Lovastatin 20 mg
-	Simvastatin 20-40 mg	Simvastatin 10 mg
-	Pravastatin 40-80 mg	Pravastatin 10-20 mg
-	Fluvastatin 80 mg	Fluvastatin 20-40 mg
-	Pitavastatin 2-4 mg	Pitavastatin 1 mg
Atorvastatin 40-80 mg	Atorvastatin 10-20 mg	
Rosuvastatin 20-40 mg	Rosuvastatin 5-10 mg	

Statins

Non-lipid effects of statins (pleiotropic effects)

- potentially relevant for the prevention of CVD
  - anti-inflammatory and antioxidant effects
  - reduced platelet aggregation

Therapeutic uses

- as **first-line** therapy for hypercholesterolemia
  - primary and secondary prevention of ASCVD
- use alone or with other lipid lowering drugs
- statins with $t_{1/2} \leq 4$ hours should be taken in the evening

Precautions: patients with renal impairment

Contraindications: pregnancy and lactation (Category X)

: continuation may be considered in patients at very high cardiovascular risk (FDA update 2021)

(Clay F.Semenkovich, Anne C. Goldberg and Ira J. Goldberg. Williams Textbook of Endocrinology, Chapter 37 , 1660-1700.)

Statins

Adverse effects

- **Common side effects:** abdominal pain, constipation, flatulence, nausea, headache, fatigue, diarrhea
- **Statin-associated muscle symptoms (SAMS)**
  - **Myopathy:** major adverse effect
    - myalgia, myositis, rhabdomyolysis
    - raised serum creatine kinase (CK) concentration → muscle damage
    - Predisposing factors : high dose and plasma concentrations
      : drug interactions
      : advanced age (> 80 years)
      : low BMI
      : hepatic or renal dysfunction
      : untreated hypothyroidism
      : genetic variation in OATP1B1 protein (*SLCO1B1*)

Statins

Adverse effects

- **Statin-induced necrotizing autoimmune myopathy (SINAM)**
  - muscle weakness + markedly ↑ CK+ anti-HMGCR antibody
  - muscle necrosis
  - Management - discontinued statins (not remit)
    - treated with immunosuppressive medications
- **Hepatotoxicity** (rare and unpredictable)
- **New-onset diabetes mellitus (DM)**
  : dose-related effect
  : individual DM risk factor– overweight, fasting blood glucose ≥ 100 mg/dL, HbA1c > 6

Statins

Drug interactions

- Mechanisms of drug interaction
  - hepatic transporter
  - CYP enzyme: 3A4, 2C9

OATP1B1 inhibitors	Gemfibrozil	increase plasma statins level
CYP3A4 inhibitors	Erythromycin, clarithromycin, itraconazole, ketoconazole, verapamil, diltiazem, HIV protease inhibitors	increase plasma statins level
CYP3A4 inducers	rifampin, carbamazepine, phenytoin	decrease plasma statins level
CYP2C9 inhibitors	Metronidazole, amiodarone, fluconazole	increase plasma statins level

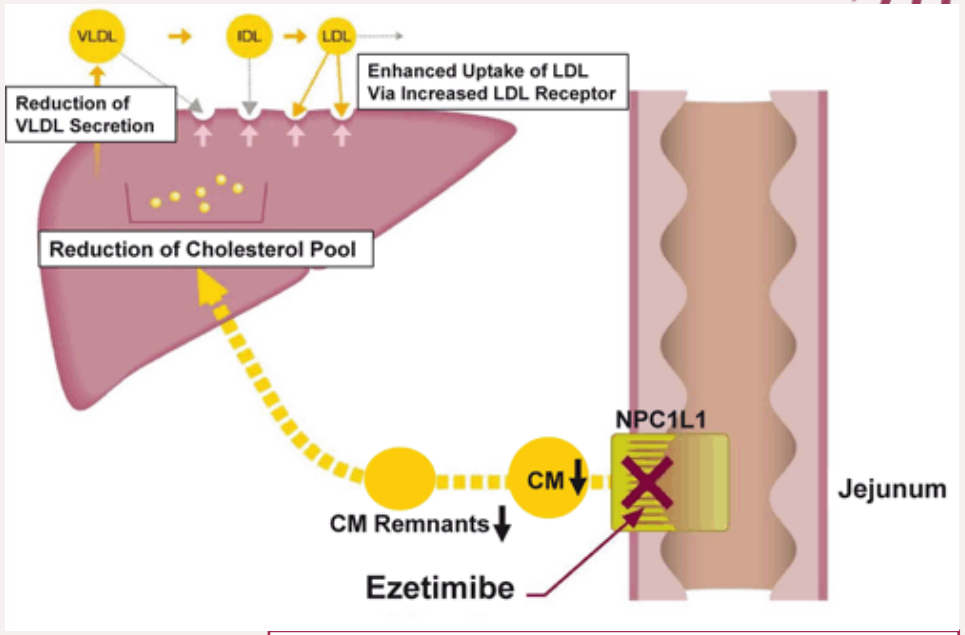
(Circulation. 2016;134:e468-e495.)

Cholesterol absorption inhibitors

- Ezetimibe

Mechanism of action

- reduce intestinal absorption of cholesterol by inhibiting NPCL1 transporter ⟶ reduce the amount of cholesterol delivered to the liver ⟶ upregulate LDL receptor expression



NPC1L1: Niemann-Pick C1-Like1

Pharmacokinetics

- conjugate to glucuronide and undergo enterohepatic circulation ⟶ long half-life (22-24 h)
- excretion: feces 70% and urine 10%

(Dyslipidemias. Contemporary Endocrinology. 2015: https://doi.org/10.1007/978-1-60761-424-1_28)

Cholesterol absorption inhibitors

Therapeutic uses

- Combination therapy (+ statins): additive lower LDL-c (by an additional 15-20%)
- Monotherapy in statin intolerant patients (lowers LDL-c by 14-25%)

Adverse effects

- diarrhea, abdominal pain
- headache, fatigue
- myositis (rare)

Contraindications

- ezetimibe: not recommended in pregnancy (preg. cat. c)
- combination tablet (with statin): pregnancy and lactation

PCSK9 inhibitors

- Pro-protein convertase subtilisin/kexin type 9 enzyme (PCSK9)
  - enhance degradation of LDL receptors in the liver

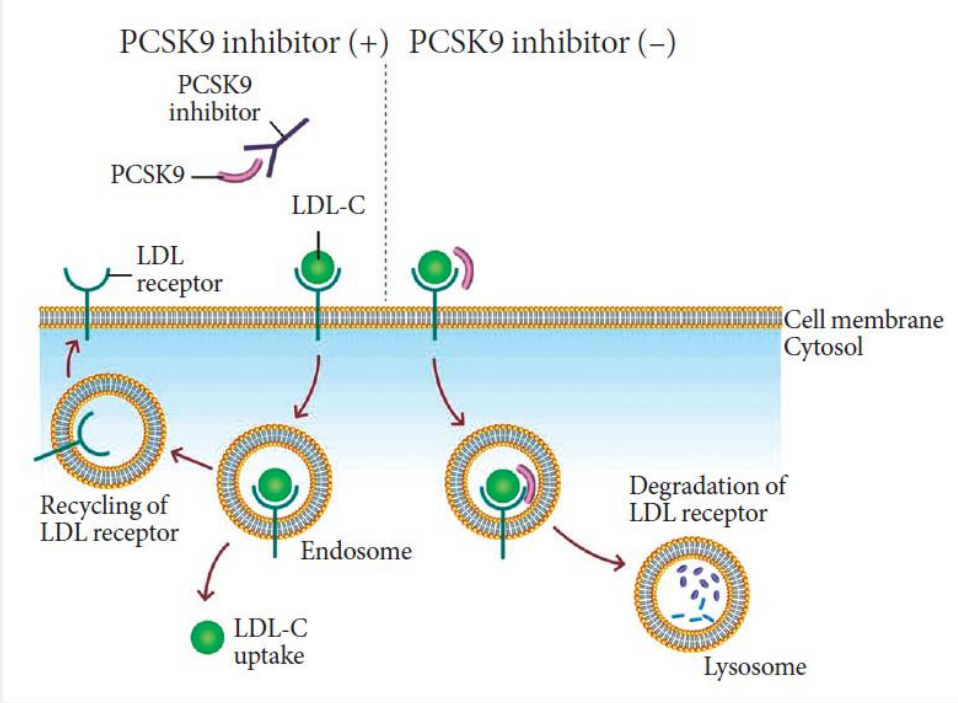
- **Human monoclonal antibody (mAb)**
  - Alirocumab, Evolocumab (SC)

Mechanism of action

- bind free PCSK9 ⟶ block the binding of PCSK9-LDL receptor ⟶ reduce degradation of the LDL receptor ⟶ ↑ liver clearance of LDL-C

Pharmacokinetics

- Metabolism: proteolysis
  : very long half-lives (11-20 days)



(Diabetes Metab J 2015;39: 87-94)

PCSK9 inhibitors

- **Human monoclonal antibody (mAb)**
  - Alirocumab, Evolocumab (SC)

Administration: every 2 weeks or once monthly (depend on dose)

Adverse effects

- neurocognitive impairment (<1%)
- injection site reactions (<10%)
- slightly increased risk of infections (UTI, URI)
- arthralgia, back pain

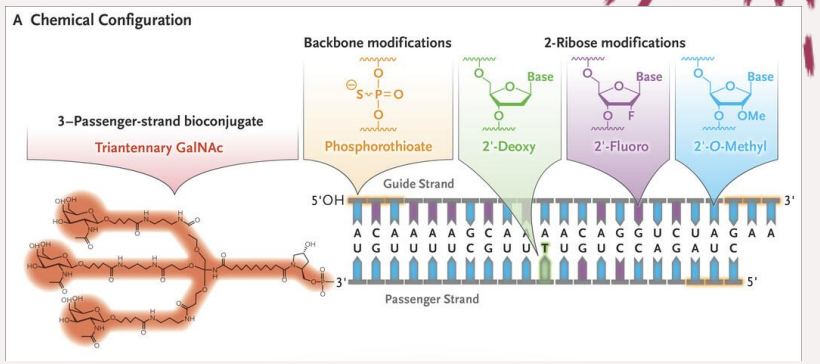
➢ should not be used in pregnancy (potential to cross placenta) and lactation

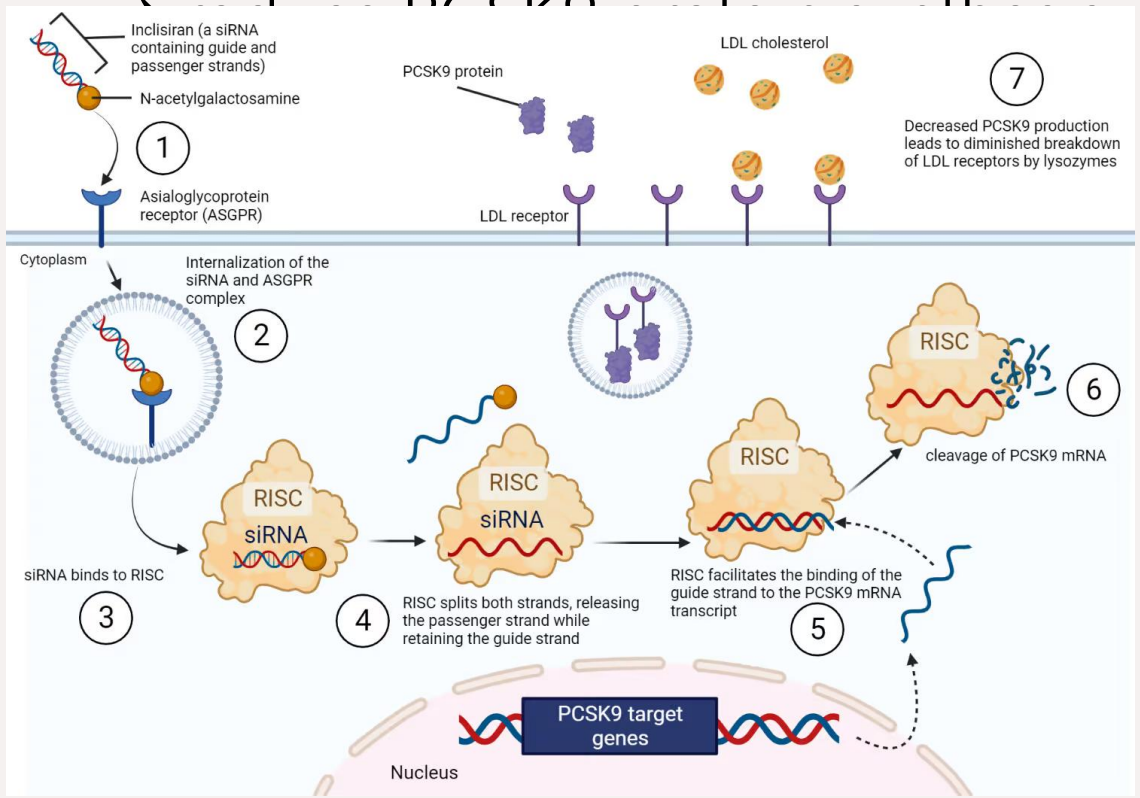
PCSK9 inhibitors

- **Small interfering ribonucleic acid (SiRNA)**
  - Inclisiran

Mechanism of action

- bind PCSK9 mRNA ⟶ cleave PCSK9 mRNA





RNA-induced silencing complex (RISC)

(N Engl J Med. 2017;376(1):4-7, https://myendoconsult.com/learn/mechanism-of-action-of-inclisiran-leqvio/#google_vignette)

PCSK9 inhibitors

- **Inclisiran**

  Pharmacokinetics: very short plasma half-life, but a very prolonged duration of action due to resistance to degradation once inside cells

  Administration: a single SC injection initially, then after 3 months, and every 6 months thereafter

  Adverse effects

  - injection-site reactions
  - arthralgia
  - urinary tract infection, bronchitis
  - diarrhea

Therapeutic use of PCSK9 inhibitors

- use to further lower LDL-C levels (below 70 mg/dl) in high-risk patients on maximally tolerated oral drugs such as statins and/or ezetimibe

Fibric acid derivatives (Fibrates)

- Clofibrate (prototype), gemfibrozil, fenofibrate, bezafibrate, pemafibrate

Mechanism of action

- Activate peroxisome proliferator-activated receptor-α (PPARα)

➢ **Reduce TG** (30-50%)

⇨ stimulate FA oxidation in liver & muscle

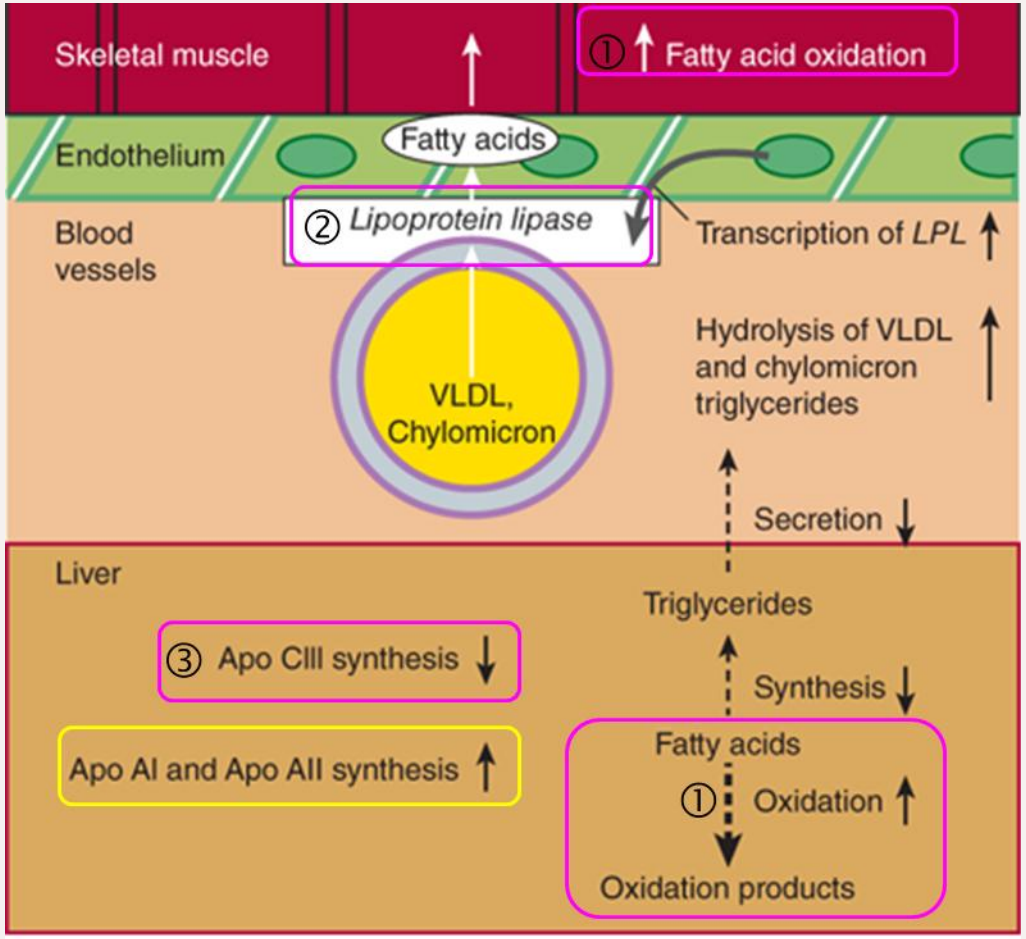
⇨ increase LPL synthesis ⟶ ↑ TG-rich lipoproteins clearance

⇨ reduce expression of apoC-III (inhibitor of lipolysis) ⟶ ↑ VLDL-c clearance

- **Increase HDL** (10-20%)

⇨ stimulate apoA-I and apoA-II expression ⟶ ↑ HDL-c levels

⇨ effectiveness: fenofibrate > gemfibrozil



Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology
www.accesspharmacy.com
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Fibric acid derivatives (Fibrates)

Pharmacokinetics

- absorbed rapidly and efficiently (>90%) when given with a meal
- half-life: fenofibrate 20 hr, gemfibrozil (1.5 hr)
- excretion: urine (adjusted dose in renal failure pt.)

Therapeutic uses

- Hypertriglyceridemia
- Mixed dyslipidemia

Fibric acid derivatives (Fibrates)

Adverse effects

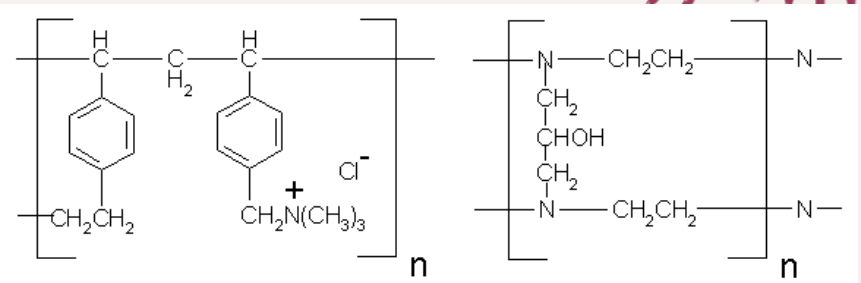
- **Myopathy**
  : occurs particularly in patients with renal impairment
  : gemfibrozil + statin ⟶ ↑ risk
    : gemfibrozil inhibits OATP1B1 and glucuronidation of statins.
- increase risk of cholesterol gallstone
- GI disturbance
- ↑ liver transaminase (minor)

Contraindications

- children and pregnant women (Pregnancy cat. c)
- severe hepatic and renal dysfunction

Bile acid-binding resins

- Bile acid sequestrants (BAS)
- cholestyramine, colestipol, colesevelam
- insoluble, nonabsorbable polymers



Mechanism of action

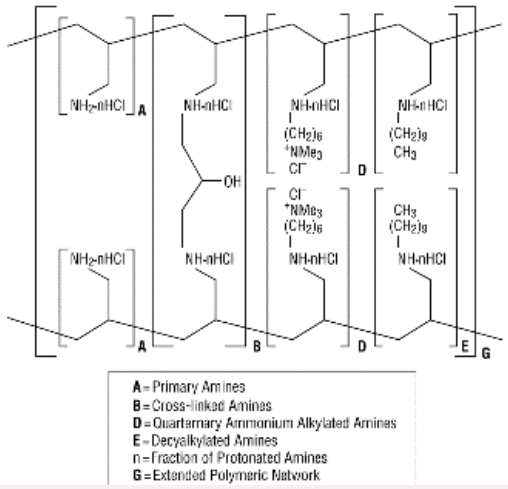
- bind bile acids and increase excretion in stool (3-10 folds)

  ➔ hepatic bile acid pool depletion

  ➔ enhance conversion of cholesterol to bile acid

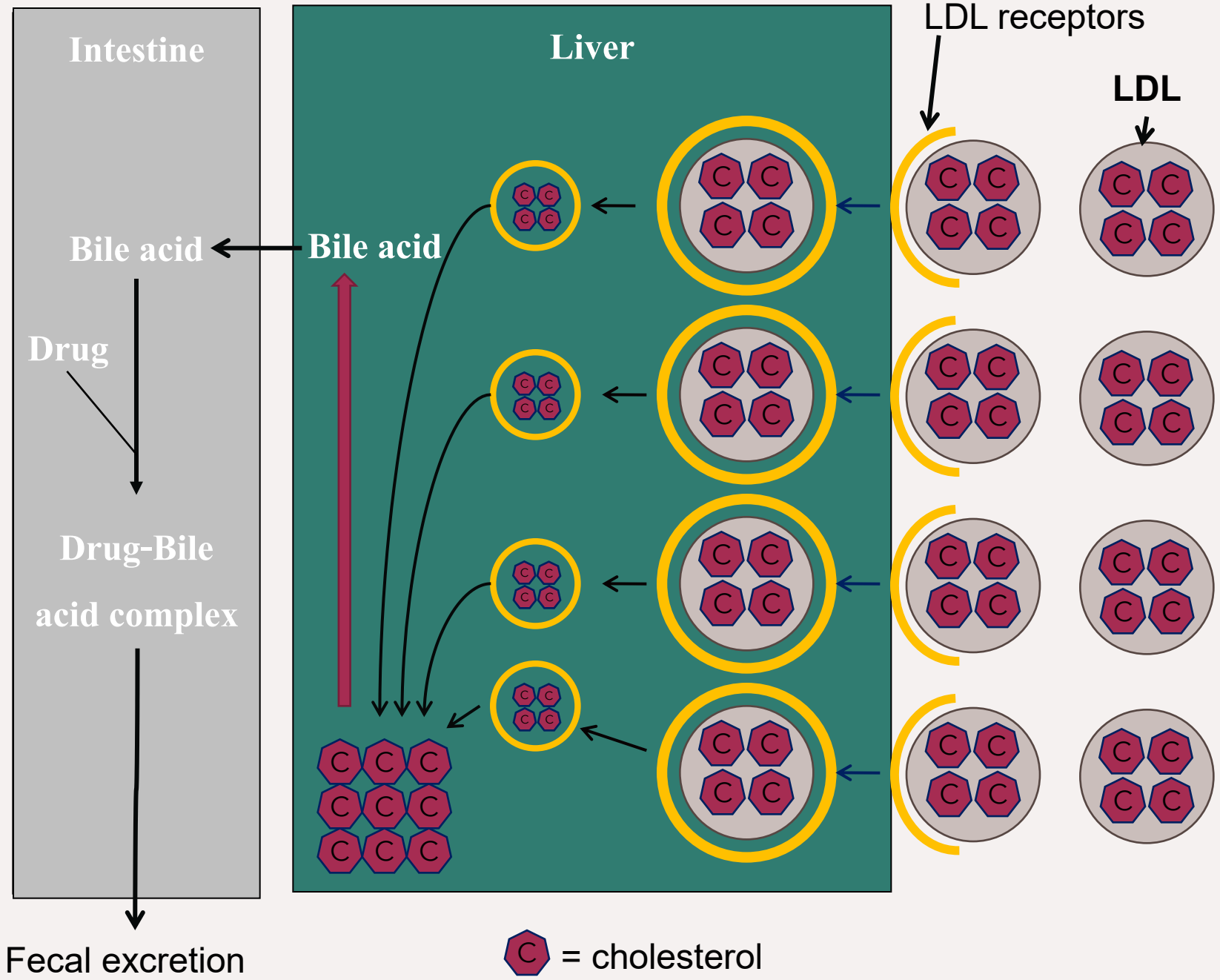
  ➔ intracellular cholesterol concentration decrease

  ➔ increase hepatic uptake of LDL by stimulating production of LDL receptor



Colesevelam

Mechanism of action



24

Bile acid-binding resins

Pharmacokinetics

- neither absorbed nor metabolized
- totally excreted in feces

Effects on lipoprotein levels

- LDL reduction
  : dose dependent (~ 20-25% reduction at maximal dose)
  : combination with statins or niacin: ↓LDL 40-60%
- may increase TG transiently⟶ contraindicate in severe hypertriglyceridemia (cholestyramine and colestipol)

Bile acid-binding resins

Adverse effects

- GI disturbances: bloating, dyspepsia, constipation, heart burn and diarrhea

Drug interactions

- interfere absorption of many drugs, fat-soluble vitamins
  : taken other drugs 1 hour before or 3-4 hours after a dose of resins

➢ Pregnancy category: B

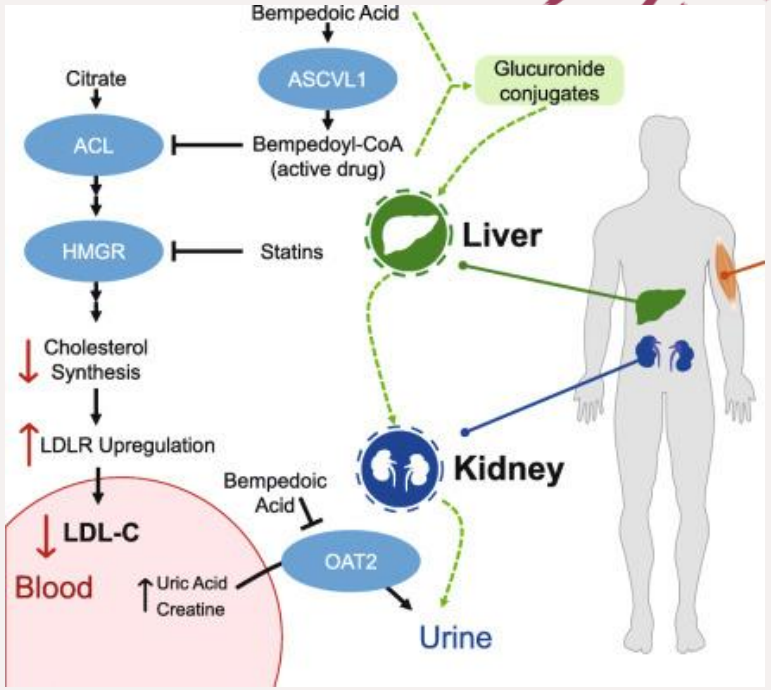
ATP-Citrate Lyase Inhibitor

- Bempedoic acid

Mechanism of action

- block ATP-citrate lyase (ACL) ⟶ inhibit cholesterol synthesis

Administration: oral



Pharmacokinetics

- Bempedoic acid = prodrug
- Metabolism and elimination
  - metabolized by ACSVL1 to bempedoyl CoA (active metabolite)
  - inactivated by UGT2B7 to glucuronide conjugates
  - half-life 15-24 hours
  - Excretion: urine (70%) by the renal transporter OAT2

(ACSVL1: very long-chain acyl-CoA synthetase-1)

(Cardiovasc Drugs Ther. 2021; 35(4): 853-864.)

ATP-Citrate Lyase Inhibitor

Therapeutic uses

- HeFH or established ASCVD who need additional LDL-C lowering
- Monotherapy: 15-25% reduction in LDL-C
- Combination uses: + ezetimibe → 30% reduction
  : triple therapy with a statin → 65% reduction

Adverse effects

- Anemia
- Hyperuricemia and gout
  - due to competition between uric acid and the bempedoic acid metabolite for the OAT2 transporter
- Tendon rupture

Drug interactions

- Simvastatin (> 20 mg) and pravastatin (>40 mg) → may increase the risk of myopathy

Contraindications: Pregnancy and lactation

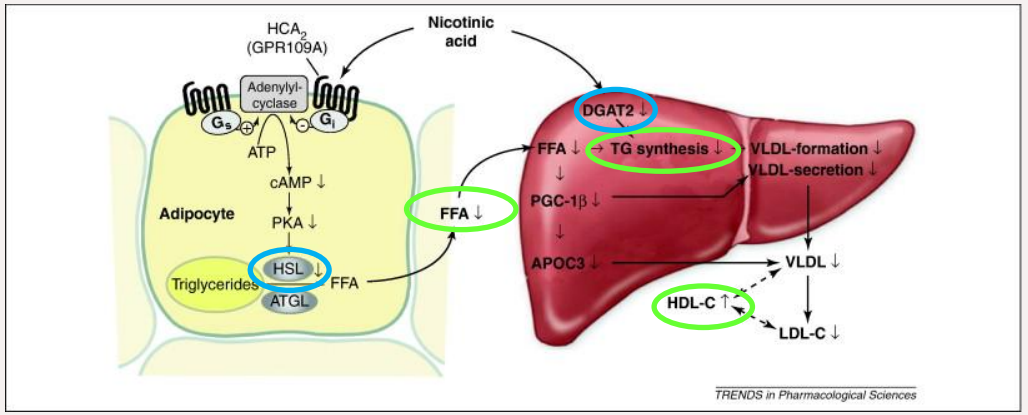
Niacin (Nicotinic acid)

Mechanism of action

Adipose tissue

- inhibit hormone-sensitive lipase (HSL)
  - via niacin receptor (HCA2)

  ⇨ inhibit lipolysis of TG → reduce transport of FFA to the liver and decrease hepatic TG synthesis



Liver

- inhibit activity of DGAT2, a key enzyme in TG synthesis

  ⇨ reduce TG synthesis and VLDL & LDL secretion

- decrease the hepatic clearance of apo A-I in HDL

  ⇨ increase HDL levels

HCA2: hydroxycarboxylic acid, PKA: protein kinase A, **HSL: hormone-sensitive lipase**, ATGL: adipocyte triglyceride lipase, FFA: free fatty acid, **DGAT2: diacylglycerol acyltransferase 2**, PGC1β: PPARγ coactivator-1β, APOC3: apolipoproteinC3),

Niacin

Effect on plasma lipoprotein levels

- ↑ HDL 30-40%, ↓ TG 35-45%
- ↓ LDL 25%

Therapeutic uses: Hypertriglyceridemia

Adverse effects

- Flushing and pruritus
  : face and upper trunk
  : prostaglandin mediated (PGD_2, PGE_2)
  : taking aspirin, ibuprofen 30-60 min in advance of niacin ⟶ reduce flushing
- GI disturbance: dyspepsia, N/V
- Hepatotoxicity

Contraindications: pregnancy